

SpendClear — Verified Agent Payments

Clean money. Clean identity. On-chain audit trail.

PROBLEM

Autonomous agents can initiate payments, but there is no on-chain mechanism to ensure the stablecoin has clean origination (no mixed or laundered funds), the recipient is a verified human identity (not a sybil wallet), spend limits are enforced per agent mandate, and every decision has an immutable audit trail for compliance reporting. Finance teams will not give agents a card because nobody can answer: who authorized this, and what stopped it going further?

SOLUTION

SpendClear is an on-chain payment governor that sits between an agent and a payment. Before any ERC-20 transfer executes, it enforces six deny-first gates: (1) caller is the authorized agent, (2) mandate is active and not revoked, (3) recipient holds a valid CVI identity token, (4) token is CVA-verified with clean origination, (5) transfer amount passes CVA compliance screening, and (6) amount is within the single-transaction and cumulative mandate caps. Every payment attempt — approved or denied — is recorded in an immutable on-chain audit log.

CVI · CVA INTEGRATION POINTS

Cleanverse Capability	Integration in SpendClear
CVI (Verified Identity)	<code>cvi.hasIdentity(recipient)</code> — gates payments to wallets with valid, non-revoked CVI identity tokens
CVA (Verified Assets)	<code>cva.isVerified(token)</code> + <code>cva.isClean(token, amount)</code> — only stablecoins with clean origination and full traceability
Agent Skill Framework	Mandate-execution pattern: principal verification, counterparty validation, spend controls, immutable audit trails
CCP Protocol	Audit log events structured for Travel Rule data extraction and compliance reporting

DEPLOYED CHAIN

Monad Testnet (Chain ID: 10143) — RPC: testnet-rpc.monad.xyz — Explorer: testnet.monadexplorer.com. High-throughput parallel EVM execution, purpose-built for the compliance workload.

DELIVERABLES

- Smart contracts: `SpendClear.sol`, `ICVI.sol`, `ICVA.sol`, `MockCVI.sol`, `MockCVA.sol`, `MockERC20.sol`
- Interactive Next.js demo: <https://web-eosin-ten-35.vercel.app>
- Demo video: 3-min walkthrough (media/SpendClear_Demo.mp4)
- GitHub: <https://github.com/icohangar-ops/spendclear>
- Track: DeFi — Compliant DeFi

PRIOR WORK DISCLOSURE

SpendClear is new, built during the Cleanverse Build hackathon (Aug 8-9, 2026). It draws on architectural patterns from MetaboSpend (agent spend governor, 78 tests, MIT, built Jul 30 - Aug 2, 2026). No code is shared — only the conceptual model of deny-first gates upstream of payment execution.